

LAB No. 15

File Permission and Ownership

Objective

To understand and manage **file permissions, ownership**, and **default permission settings (umask)** in Linux using administrative commands.

Learning Outcomes

After completing this lab, students will be able to:

1. Interpret Linux file permission symbols and numeric modes.
2. Change file and directory permissions using **chmod**.
3. Modify file ownership and group using **chown** and **chgrp**.
4. Configure default file permissions using **umask**.
5. Understand security implications of permissions.

Lab Requirements

- **OS:** Ubuntu / CentOS / Rocky Linux
- **Access:** Normal user with sudo privileges
- **Tools:** Terminal

Important Commands and Descriptions

Command	Description	Example
<code>ls -l</code>	Lists files with permissions, owner, group, and size	<code>ls -l /home/student</code>
<code>chmod</code>	Changes file or directory permissions	<code>chmod 755 file1.txt</code>
<code>chown</code>	Changes file ownership	<code>sudo chown user1 file1.txt</code>
<code>chgrp</code>	Changes file group ownership	<code>sudo chgrp developers file1.txt</code>
<code>umask</code>	Sets default permission mask for new files/directories	<code>umask 022</code>

Task 1: View Current Permissions

1. Open terminal and create files:

```
touch file1.txt  
mkdir myfolder
```

2. View permissions:

```
ls -l
```

Output Example:

```
-rw-r--r-- 1 student student 0 Oct 16 11:32 file1.txt  
drwxr-xr-x 2 student student 4096 Oct 16 11:32 myfolder
```

- r = Read
- w = Write
- x = Execute
- First 3 characters → Owner permissions
- Next 3 → Group permissions
- Last 3 → Others

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ touch file1.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mkdir myfolder
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l
total 420
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 195 Apr  2 15:04 append_to_file.sh
drwxrwxr-x 2 abdullah-sohail abdullah-sohail 4096 Apr  4 16:10 backup
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 225 Apr  2 15:11 backup_file.sh
drwxrwxr-x 2 abdullah-sohail abdullah-sohail 4096 Mar 31 17:11 bash_lab
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 445 Mar 31 17:56 calculator.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 195 Apr  2 15:07 check_file_exists.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 226 Mar 27 12:19 check_file.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 202 Mar 31 17:40 check_number.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 181 Apr  2 15:13 count_lines.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 187 Apr  2 15:02 create_file.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 207 Apr  2 15:09 delete_file.sh
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail  44 Mar 27 12:13 demo.sh
drwxr-xr-x 2 abdullah-sohail abdullah-sohail 4096 Feb 18 19:56 Desktop
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 335 Mar 31 18:04 dir_check.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 204 Mar 31 18:02 divide.sh
drwxr-xr-x 2 abdullah-sohail abdullah-sohail 4096 Feb 18 19:56 Documents
drwxr-xr-x 2 abdullah-sohail abdullah-sohail 4096 Mar 25 22:27 Downloads
drwxrwxr-x 2 abdullah-sohail abdullah-sohail 4096 Mar 31 11:38 duplicates
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail  30 Mar 31 11:36 File1.jpg.gz
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail   0 Apr 24 10:50 file1.txt
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 405 Mar 31 17:45 file_check.sh
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 340 Mar 31 17:59 file_function.sh
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail  45 Apr  4 16:42 file.txt
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail  32 Mar 31 11:41 important.txt
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail  21 Mar 31 11:41 input.txt
drwxrwxr-x 2 abdullah-sohail abdullah-sohail 4096 Mar 31 11:26 large_files
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 172 Mar 31 17:51 leap_year.sh
drwxr-xr-x 2 abdullah-sohail abdullah-sohail 4096 Feb 18 19:56 Music
```

Task 2: Change Permissions Using chmod

a) Symbolic Method:

- `chmod u+x file1.txt` **# Add execute for user**
- `chmod g-w file1.txt` **# Remove write for group**
- `chmod o+r file1.txt` **# Add read for others**

b) Check results:

```
ls -l file1.txt
```

b) Numeric Method:

Mode	Meaning	Binary	Octal
r (read)	4	100	4
w (write)	2	010	2
x (execute)	1	001	1

Example:

```
chmod 740 file1.txt
```

- Owner: 7 (rwx)
- Group: 4 (r--)
- Others: 0 (---)

Task 3: Change Ownership and Group

1. Create a test user and group:

```
sudo useradd testuser  
sudo groupadd devgroup
```

2. Assign file ownership:

```
sudo chown testuser file1.txt
```

3. Assign group ownership:

```
sudo chgrp devgroup file1.txt
```

4. Combine both:

```
sudo chown testuser:devgroup file1.txt
```

5. Verify

```
ls -l file1.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~  
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 124 Mar 27 12:31 task6.sh  
-rwxrwxr-x 1 abdullah-sohail abdullah-sohail 113 Mar 27 12:33 task7.sh  
drwxr-xr-x 2 abdullah-sohail abdullah-sohail 4096 Feb 18 19:56 Templates  
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail 15 Apr 2 15:15 test1.txt  
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail 15 Apr 2 15:16 test1.txt.bak  
drwxr-xr-x 2 abdullah-sohail abdullah-sohail 4096 Feb 18 19:56 Videos  
drwxrwxr-x 3 abdullah-sohail abdullah-sohail 4096 Mar 31 11:34 vm_lab  
drwxr-xr-x 2 root root 4096 Mar 5 12:41 webapp  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod u+x file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod g-w file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod o+r file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l file1.txt  
-rwxr--r-- 1 abdullah-sohail abdullah-sohail 0 Apr 24 10:50 file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod 740 file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l file1.txt  
-rwxr----- 1 abdullah-sohail abdullah-sohail 0 Apr 24 10:50 file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd testuser  
[sudo] password for abdullah-sohail:  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo groupadd devgroup  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo chown testuser file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l file1.txt  
-rwxr----- 1 testuser abdullah-sohail 0 Apr 24 10:50 file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo chgrp devgroup file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l file1.txt  
-rwxr----- 1 testuser devgroup 0 Apr 24 10:50 file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo chown testuser:devgroup file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l file1.txt  
-rwxr----- 1 testuser devgroup 0 Apr 24 10:50 file1.txt  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ umask  
0002  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```